CRIMEVISIONAI

Performance Testing

Folder 6 | Honest test plan and results record | Prepared 23 October 2026

Project: CrimeVisionAI — Multimodal Forensic Evidence Analysis and Investigation Support System

Document status: Project-specific draft based on the implementation information currently available. Verify placeholders and run-dependent results before formal submission.

1. Testing overview

Field	Current record
Tool	Not yet specified / no benchmark output supplied
Testing type	Performance test plan; load/stress test not evidenced
Target modules	Case/evidence API, upload, audio/image/video analysis and Q&A
Environment	Local development; reported Flask server at 127.0.0.1:5000; hardware details not supplied
Test date	To be filled after executing the benchmark
Result status	Not measured — no numerical pass/fail claims are made in this document.

2. Recommended test scenarios

ID	Scenario	Load / duration	Expected observation
PT-01	Case list and case metadata endpoint	1 user, 30 requests	Record median/p95/max latency and errors.
PT-02	Small image upload and analysis	1 user, 5 representative images	Record upload time, analysis time, memory and output correctness.
PT-03	Audio analysis with short sample	1 user, 5 representative clips	Record transcription/analysis duration and failure rate.
PT-04	Video analysis near configured cap	1 user, 1 controlled clip up to 180 seconds	Record wall time, frame count, memory and errors.
PT-05	Concurrent metadata requests	5 concurrent clients for 60 seconds	Observe contention, response latency and error rate.
PT-06	Concurrent ML analysis	2–3 jobs only after baseline; 60 seconds	Assess CPU/GPU and memory pressure; stop if resource saturation occurs.
PT-07	LLM API unavailable / rate-limited	Controlled failure injection	Confirm clear error handling and no corrupt partial records.

3. Results table — complete only after execution

Metric	Target / method	Actual value	Status
Metadata API median / p95 latency	Set after baseline	Not measured	Pending
Upload latency by file size	Record size and duration	Not measured	Pending
Audio analysis wall time	Report clip duration and hardware	Not measured	Pending
Image analysis wall time	Report image dimensions and hardware	Not measured	Pending
Video analysis wall time	Report duration, sampled frames and hardware	Not measured	Pending
Throughput	Requests per second for defined endpoint	Not measured	Pending
Error rate	Failed requests / total requests	Not measured	Pending
Peak CPU / RAM / GPU memory	Capture with monitoring tool	Not measured	Pending

4. Tools and repeatability

Suggested tools: pytest for route/helper tests; curl or Postman for endpoint smoke tests; Locust, k6 or JMeter for HTTP load tests; operating-system or Python monitoring for CPU/RAM; GPU monitoring if CUDA is used. Record Python version, model/device, OS, input size/duration, concurrency, warm-up behavior, model cache state and whether Groq calls are included.

5. Analysis guidance

Do not use generic expectations such as average response under two seconds for model inference without a baseline. Metadata routes and ML inference have different performance profiles. Report median and p95 alongside average, separate local inference from external API latency, and repeat runs. Do not infer production capacity from a single-user laptop demonstration.

6. Evidence to attach

After running tests, insert dated screenshots or raw output files, the exact command/configuration, hardware details and a concise interpretation. Until then, this document is a complete test plan but not a completed performance-test report.